5.10 Classwork: Tangent Lines and Enclosed Area — Markscheme

1. $f(x) = 4x - x^2$, tangent at $P(1, 3)$.

   (a) $f'(x) = 4 - 2x$ **(A1)**

   $f'(1) = 2$, so gradient of L is 2.

   $y - 3 = 2(x - 1) \Rightarrow y = 2x + 1$ **(AG)**

   (b) Area $= \displaystyle\int_0^1 \big((2x + 1) - (4x - x^2)\big)\,\mathrm{d}x$ **(M1)**

   $= \displaystyle\int_0^1 (x^2 - 2x + 1)\,\mathrm{d}x = \int_0^1 (x - 1)^2\,\mathrm{d}x$ **(A1)**

   $= \left[\dfrac{(x - 1)^3}{3}\right]_0^1 = 0 - \left(\dfrac{-1}{3}\right)$ **(A1)**

   $= \dfrac{1}{3}$ **(A1)**

2. $f(x) = e^x$, tangent at $P(1, e)$.

   (a) $f'(x) = e^x$ **(A1)**

   (b) $f'(1) = e$ **(M1)**

   $y - e = e(x - 1) \Rightarrow y = ex$ **(AG)**

   (c) Area $= \displaystyle\int_0^1 \big(e^x - ex\big)\,\mathrm{d}x$ **(M1)**

   $= \left[e^x - \dfrac{ex^2}{2}\right]_0^1$ **(A1)(A1)**

   $= \left(e - \dfrac{e}{2}\right) - (1 - 0) = \dfrac{e}{2} - 1$ **(A1)**

3. $f(x) = x^3$, tangent at $P(1, 1)$.

   (a) $f'(x) = 3x^2$ **(A1)**

   (b) $f'(1) = 3$ **(M1)**

   $y - 1 = 3(x - 1) \Rightarrow y = 3x - 2$ **(AG)**

   (c) $x^3 = 3x - 2$ **(M1)**

   $x^3 - 3x + 2 = 0$

$(x-1)^2(x+2)=0$ **(A1)**

$Q=(-2,\ -8)$ **(A1)**

(d) Area $= \displaystyle\int_{-2}^{1}\big(x^3-(3x-2)\big)\,\mathrm{d}x$ **(M1)**

$= \displaystyle\int_{-2}^{1}(x^3-3x+2)\,\mathrm{d}x$

$= \left[\dfrac{x^4}{4}-\dfrac{3x^2}{2}+2x\right]_{-2}^{1}$ **(A1)(A1)**

$= \left(\dfrac{1}{4}-\dfrac{3}{2}+2\right)-\left(\dfrac{16}{4}-\dfrac{12}{2}-4\right)$ **(M1)**

$= \dfrac{3}{4}-(-6)=\dfrac{27}{4}$ **(A1)**

4. $f(x)=6e^{-0.5x}$, tangent at $A(0,6)$.

(a) $A=(0,\ 6)$ **(A1)**

(b) $f'(x)=6\times(-0.5)\,e^{-0.5x}$ **(M1)**

$=-3e^{-0.5x}$ **(A1)**

(c) $f'(0)=-3$ **(A1)**

$y-6=-3(x-0)\Rightarrow y=-3x+6$ **(AG)**

(d) $0=-3x+6\Rightarrow x=2$

$B=(2,\ 0)$ **(A1)**

(e) Area $= \displaystyle\int_{0}^{2}\big(6e^{-0.5x}-(-3x+6)\big)\,\mathrm{d}x$ **(M1)**

GDC: $=1.5854\ldots$ **(A1)**

≈ 1.59 **(A1)**

Exact form: $\left[-12e^{-0.5x}+\frac{3}{2}x^2-6x\right]_0^2=(-12e^{-1}+6-12)-(-12)=6-12e^{-1}$